```
%% DEFINE VARIABLES
stream1 = 1000; %kg/hr Inlet
stream2 = 1000;

% Stream 1 composition
A1= 0.3;
B1= 0.3;
C1= 0.2;
D1= 0.2;      % MOD: added C5

% Stream 2 composition
A2= 0.3;
B2= 0.2;
C2= 0.4;
D2= 0.1;      % MOD: added C5

% Stream A (Stream 3)
A3= 0.6;
B3= 0.2;
D3= 0.2;      % MOD: added C5

% Stream C (Stream 5)
B5 = 0.1;
A5 = 0.7;
D5 = 0.2;     % MOD: C5 leaving through C

% Stream D (Stream 6)
A6 = 0.2;
B6 = 0.8;

% Stream E (Stream 7)
B7= 0.3;
C7 = 0.7;

%% Stream G (Recycle) NEW
Stream8 = 500;

BG = 0.6;
CG = 0.3;
DG = 0.1;

%% Solving for streams C, D, E
syms C D E

C3_Balance = stream1*B1 + stream2*B2 == C*B5 + D*B6 + E*B7;
C2_Balance = stream1*A1 + stream2*A2 == C*A5 + D*A6;
C4_Balance = stream1*C1 + stream2*C2 == E*C7;

soln = solve(C3_Balance, C2_Balance, C4_Balance, C, D, E);
C = eval(soln.C);
D = eval(soln.D);
E = eval(soln.E);

% Solve for stream B
B = (((A3*A5*C)/A3)-(A3*C))/.2;

% Stream A
A = B + C;
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```

%% Stream G added
G = Stream8;     % MOD

%% Mixer (F = D + E + G)
F = D + E + G;   % MOD: added +G

%% Solve for F composition
syms C2_F C3_F C4_F C5_F

eq1 = F*C2_F == D*A6;
eq2 = F*C3_F == D*B6 + E*B7 + G*BG;   % MOD: +G*BG
eq3 = F*C4_F == E*C7 + G*CG;          % MOD: +G*CG
eq4 = C2_F + C3_F + C4_F + C5_F == 1; % MOD: added C5

solnF = solve(eq1,eq2,eq3,eq4,C2_F,C3_F,C4_F,C5_F);

C2_F = eval(solnF.C2_F);
C3_F = eval(solnF.C3_F);
C4_F = eval(solnF.C4_F);
C5_F = eval(solnF.C5_F);              % MOD: C5 in F

%% Display F composition
FComposition= {'C2';'C3'; 'C4'; 'C5'};
WtFrac= [C2_F,C3_F,C4_F,C5_F];

fprintf('Stream F Mass Fraction  \n');
fprintf('-------------------------------\n');
for i = 1:length(FComposition)
    fprintf('%s\t\t| %.2f\n', FComposition{i}, WtFrac(i));
end

%% Display flowrates
Stream={'A'; 'B'; 'C'; 'D'; 'E';'F';'G'}; % MOD: added G
FlowRate= [A; B; C; D; E; F; G];

fprintf('\nStream\t\t| Flow Rate (kg/hr)\n');
fprintf('------------------------------\n');
for i = 1:length(Stream)
    fprintf('%s\t\t| %.0f\n', Stream{i}, FlowRate(i));
end

%% Checks
if C+D+E == stream1+stream2
    fprintf('\nMass is conserved (splitter)')
end

if abs((D+E+G)-F) < 1e-6  % MOD: includes G
    fprintf('\nMass is conserved (mixer)\n')
end

% Stream F Mass Fraction
% -------------------------------
% C2        | 0.03
% C3        | 0.46
% C4        | 0.48
% C5        | 0.03
%
```

```
% Stream        | Flow Rate (kg/hr)
% -------------------------------
% A        | 1198
% B        | 399
% C        | 799
% D        | 204
% E        | 857
% F        | 1561
% G        | 500
%
% Mass is conserved (mixer)
```